

ERTH 467 – Time Series Analysis and 1D Signal Processing (F2026)

Lab Assignment #3

In this lab assignment, you will develop a code to perform Wiener deconvolution in the time domain. The data for this assignment is provided in the files

“erth467_lab3_data1.mat”,
“erth467_lab3_data2.mat”,
“erth467_lab3_data1.txt”,
“erth467_lab3_data2.txt” and
“erth467_lab3_wavelet.txt”

on D2L (the files contain the same data; use the .mat file if you are coding in MATLAB and the .txt files otherwise). Based on your results, you will write a brief scientific report (see report format and rubric under Lab Information on D2L) discussing your results. You will submit the assignment to the Dropbox on D2L as a compressed (.zip) file entitled “ERTH467_Lab3_Name.zip” containing a copy of your report in .pdf format and a copy of all code used to generate your report results. In the file name, replace Name with your surname.

The data files contain both a noise-free data set (erth467_lab3_data1.mat), for which the wavelet is known (erth467_lab3_wavelet.mat), and a noisy data set (erth467_lab3_data2.mat). Please note that these data sets do not correspond to the same subsurface model. The variables “data1” and “t1” contain the data and measurement times for the noise-free data set, the variables “data2” and “t2” contain the data and measurement times for the noisy data set, and the variables “w” and “tw” contain the wavelet and associated time for the noise-free data set. Both data sets were created using minimum-phase wavelets.

1. Write a function that takes an amplitude spectrum and frequency vector as inputs and outputs the corresponding minimum-phase time-domain wavelet and associated times. To check your function, try applying it to the spectrum of the minimum-phase wavelet in “w”. The MATLAB function `hilbert` will be useful here. Remember that the Hilbert transform of x in MATLAB is `imag(hilbert(x))`.
2. Write a function to perform zero-lag Wiener deconvolution. Use your function to deconvolve both data sets. How do the results compare? How did you decide on how large to make the stabilization term in each case? Create a set of figures comparing the deconvolution results to the original data.
3. Write a function that allows time lags in Wiener deconvolution. Use your function to deconvolve both data sets. How do the results compare to each other? How do they compare to the results from part 2? How did you create the wavelet estimate for the data set without a known wavelet? How did you choose the time lags? Did the size of the stability term that was needed change? Create a set of figures comparing the deconvolution results to the original data and the results from part 2.

Write a brief report on your results following the format posted on D2L. The report and code (together in a compressed file) should be submitted to the Dropbox on D2L by December 1, 2026.